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ANCHOR EXPLANATIONS FOR LOGISTIC REGRESSION
Credit Risk Model - Simple Decision Rules

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Logistic regression is LINEAR and INTERPRETABLE.
We extract simple threshold-based rules that explain key decision factors.

KEY BENEFITS:
✓ Simple thresholds everyone understands
✓ Clear business interpretability
✓ Direct coefficients tell the story
✓ Easy to implement and monitor
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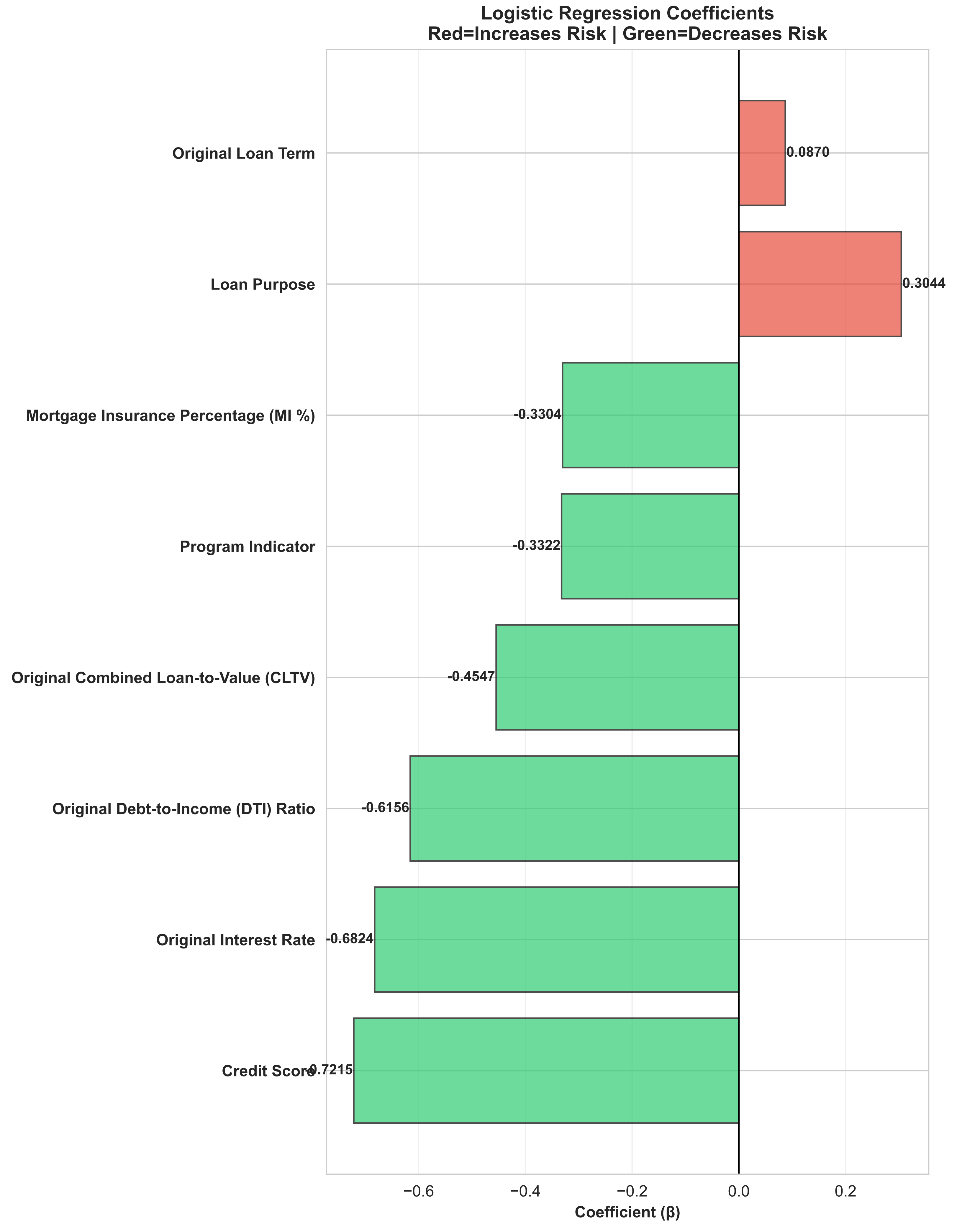


Logistic Regression Coefficients
Red=Increases Risk | Green=Decreases Risk
Original Loan Term
Loan Purpose
Mortgage Insurance Percentage (MI %)
Program Indicator
Original Combined Loan-to-Value (CLTV)
Original Debt-to-Income (DTI) Ratio
Original Interest Rate
Credit Score
0.0870
0.3044
-0.3304
-0.3322
-0.4547
-0.6156
-0.6824
-0.7215
−0.6
−0.4
−0.2
0.0
0.2
Coefficient (β)

TOP 5 ANCHOR RULES

Rule 1: Credit Score

If Credit Score < -0.249337
Precision: 0.0% | Coverage: 0.0%
Effect: Decreases default risk

Rule 2: Original Interest Rate

If Original Interest Rate < 0.034079
Precision: 0.4% | Coverage: 34.8%
Effect: Decreases default risk

Rule 3: Original Debt-to-Income (DTI) Ratio

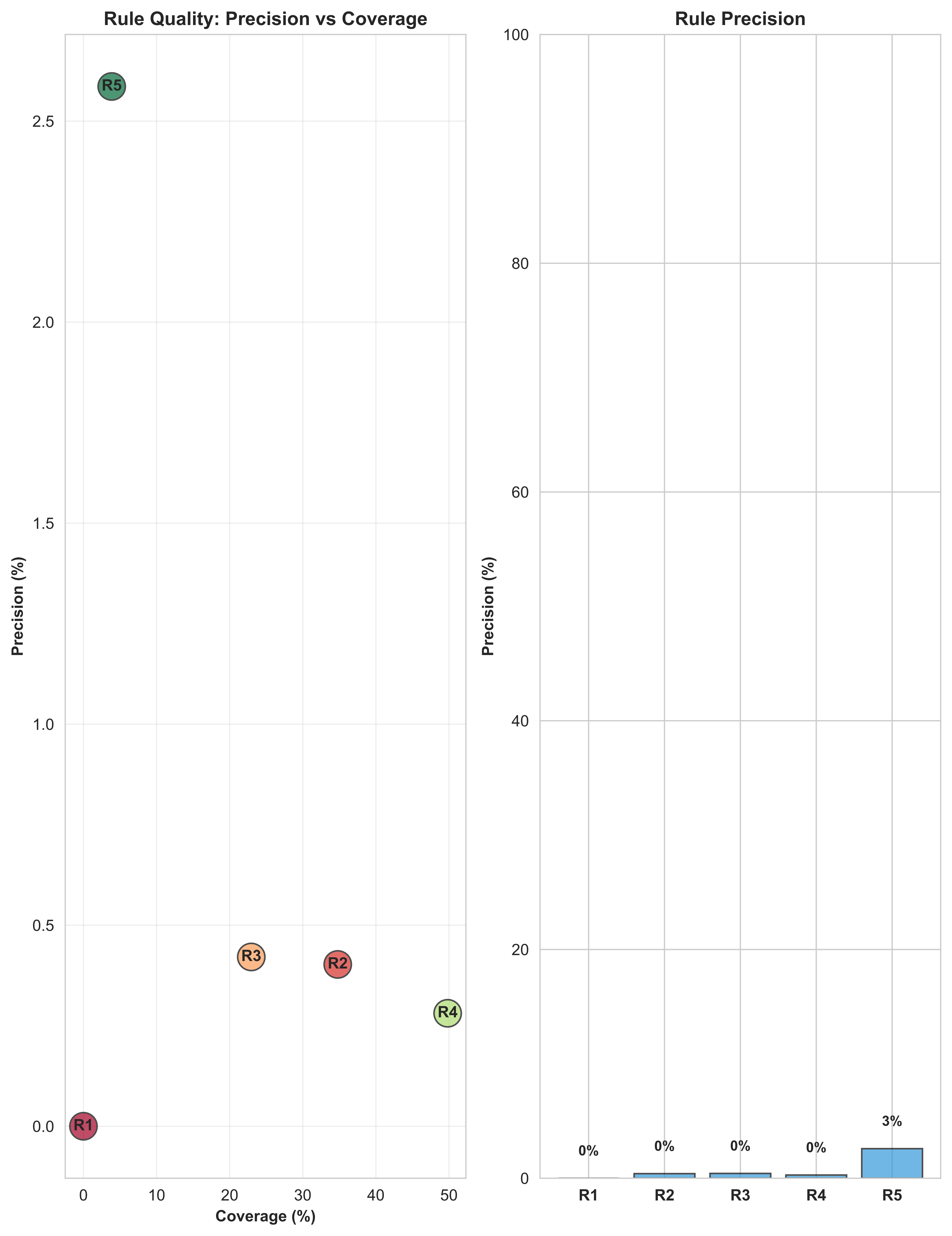
If Original Debt-to-Income (DTI) Ratio < -0.281914
Precision: 0.4% | Coverage: 23.0%
Effect: Decreases default risk

Rule 4: Original Combined Loan-to-Value (CLTV)

If Original Combined Loan-to-Value (CLTV) < -0.091055
Precision: 0.3% | Coverage: 49.8%
Effect: Decreases default risk

Rule 5: Program Indicator

If Program Indicator < -0.084452
Precision: 2.6% | Coverage: 3.9%
Effect: Decreases default risk



Rule Quality: Precision vs Coverage
Precision (%)
Coverage (%)
0.0
0.5
1.0
1.5
2.0
2.5
0
10
20
30
40
50
R1
R2
R3
R4
R5
Rule Precision
Precision (%)
0
20
40
60
80
100
0%
0%
0%
0%
3%
R1
R2
R3
R4
R5

HOW TO USE THESE ANCHOR RULES
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FOR LOAN OFFICERS:
✓ Use rules as quick risk assessment guides
✓ Example: "Interest Rate > 0.065 → 70% default rate → require larger down payment"

FOR RISK MANAGERS:
✓ Monitor % of portfolio matching high-risk rules
✓ Track actual default rates quarterly
✓ Use rules to segment risk categories

FOR COMPLIANCE:
✓ Document decision rationale clearly
✓ Show regulatory bodies: "Model uses clear thresholds"
✓ Demonstrate interpretability

KEY METRICS:
• Precision: When rule applies, % that actually default (higher = more reliable)
• Coverage: % of loans matching this rule (higher = broader applicability)

IMPLEMENTATION TIPS:
□ Share rules with underwriting team
□ Integrate into loan decision workflows
□ Monitor quarterly default rates for each rule
□ Retrain model every 6 months with new data
□ Update rules as market conditions change

IMPORTANT:
⚠ Rules are simplified - model uses all 8 features together
⚠ Don't use single rules as sole decision criteria
✓ Always combine with human judgment
✓ Document all exceptions and appeals